Notes on Multivariable Calculus

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1 Vectors

1.1 Basics

Let u, v be vectors in $\mathbb{R}^n$. They will have n components, and let them be as follows, where $u_i, v_i \in \mathbb{R}$ (scalars):

$$u = (u_1, u_2, \ldots, u_n)$$
$$v = (v_1, v_2, \ldots, v_n)$$

We can define the addition operation and the magnitude of each vector as:

$$u + v = (u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n)$$
$$\|u\| = \sqrt{u_1^2 + u_2^2 + \ldots + u_n^2} = \sqrt{\sum_{i=1}^{n} u_i}$$

Since scalar addition is commutative, so is vector addition. Also, the subtraction operation is defined as:

$$u - v = u + (-v) = (u_1 - v_1, u_2 - v_2, \ldots, u_n - v_n)$$

If we have two points A and B, the vector AB from A to B is given by $AB = B - A$. If we placed the tail of the vector at A, then the head will point to B.

Two vectors are parallel if and only if they are scalar multiples of each other; that is, there exists a scalar $a \in \mathbb{R}$ such that $u = av$.

1.2 Polar Form

For a vector $u \in \mathbb{R}^2$, we can write u in polar form. Any vector can be specified with an angle from the x axis and a magnitude. Let the angle from the x axis be θ and the magnitude, the distance to the origin, be r. To convert from cartesian to polar:

$$u = \begin{bmatrix} u_x \\ u_y \end{bmatrix} \implies \begin{cases} \theta = \arctan\left(\dfrac{u_y}{u_x}\right) \\ r = \|u\| = \sqrt{u_x^2 + u_y^2} \end{cases}$$

To convert from polar to cartesian:

$$u = \begin{bmatrix} r\cos\theta \\ r\sin\theta \end{bmatrix} = r \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

Note that since $\sin^2\theta + \cos^2\theta = 1$ by the Pythagorean identity, the magnitude of u is r.

1.3 Unit Vectors

A unit vector is a vector of magnitude 1. For a vector $u \in \mathbb{R}^n$, a unit vector with the same direction is $u/\|u\|$.

1.4 Dot Product

For vectors $u = (u_1, \ldots, u_n) \in \mathbb{R}^n$ and $v = (v_1, \ldots, v_n) \in \mathbb{R}^n$, the cross product of the two vectors $u \cdot v$ is:

$$u \cdot v = u_1 v_1 + u_2 v_2 + \ldots + u_n v_n = \sum_{i=1}^{n} u_i v_i$$

Since scalar multiplication is commutative, so is the dot product: $u \cdot v = v \cdot u$.

Alternatively, let θ be the angle between vectors u and v. Then:

$$u \cdot v = \|u\| \|v\| \cos\theta$$

1.5 Orthogonality

Two vectors u and v are orthogonal if and only if $u \cdot v = 0$.

1.6 Standard Bases

The standard basis vectors are as follows:

$$\hat{i} = (1, 0, 0)$$
$$\hat{j} = (0, 1, 0)$$
$$\hat{k} = (0, 0, 1)$$

These vectors span the $\mathbb{R}^3$ space.

1.7 Cross Product

The cross product only exists between vectors in three dimensions. Let $u = (u_x, u_y, u_z) \in \mathbb{R}^3$ and $v = (v_x, v_y, v_z) \in \mathbb{R}^3$. The cross product $u \times v$ has the following properties:

- $u \times v$ is a vector in $\mathbb{R}^3$.
- $u \times v$ is orthogonal to both u and v. If u and v are linearly independent and span a plane, $u \times v$ is orthogonal to that plane as well.
- $u \times v = 0$ if and only if u and v are parallel/linearly dependent.
- The magnitude of $u \times v$ equals the area of the parallelogram formed by u and v: $\|u \times v\| = \|u\| \|v\| \sin\theta$

The formula for $u \times v$ can be written concisely with a determinant, and expanding by cofactors:

$$u \times v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix} = \hat{i} \begin{vmatrix} u_y & u_z \\ v_y & v_z \end{vmatrix} - \hat{j} \begin{vmatrix} u_x & u_z \\ v_x & v_z \end{vmatrix} + \hat{k} \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$$

The operation has the following properties:

- Anticommutativity: $u \times v = -(v \times u)$
- Distributive over addition: $(u + v) \times w = u \times w + v \times w$ and $u \times (v + w) = u \times v + u \times w$.
- Multiplication by a scalar: $(cu) \times v = c(u \times v) = u \times (cv)$ where $c \in \mathbb{R}$.

The scalar triple product is the determinant of the square matrix with those vectors as its rows (or columns):

$$u \cdot (v \times w) = (v \times w) \cdot u = \begin{vmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix}$$

From the determinant, it follows that the scalar triple product of u, v, w is the volume of the parallelopiped with u, v, w as its edges.

1.8 Projection

The orthogonal projection of u onto v, written $\text{proj}_v u$, is the vector parallel to v such that the error between the projection and u is minimized.

Let p be the orthogonal projection $\text{proj}_v u$. The error of the project is $e = u - p$. The error is minimized when e is orthogonal to v. The formula for orthogonal projection is:

$$\text{proj}_v u = \frac{u \cdot v}{v \cdot v} v = \frac{u \cdot v}{\|v\|^2} v$$

The proof is left as an exercise to the reader.

2 Lines

A line in 3D space can be represented in three ways. In each subsection below, we specify a line passing through (x_0, y_0, z_0) with slopes a, b, c.

2.1 Parametric

This parametric system with parameter t describes the line we want:

$$\begin{cases} x(t) = at + x_0 \\ y(t) = bt + y_0 \\ z(t) = ct + z_0 \end{cases}$$

2.2 Vector

A vector valued function $f : \mathbb{R} \to \mathbb{R}^n$. It takes a scalar and returns a vector: $f(t) \in \mathbb{R}^n, t \in \mathbb{R}$.

From the parametric equation of a line, we notice that for parameter t, this point lies on the line. We can define a vector valued function r with parameter t. This is the vector equation for a line.

$$r(t) = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} at + x_0 \\ bt + y_0 \\ ct + z_0 \end{bmatrix}$$

2.3 Symmetric

Recall the parametric system for a line. We solve for the parameter t in terms of x, y, z values:

$$\begin{cases} x(t) = at + x_0 \\ y(t) = bt + y_0 \\ z(t) = ct + z_0 \end{cases} \implies \begin{cases} t = \dfrac{x - x_0}{a} \\ t = \dfrac{y - y_0}{b} \\ t = \dfrac{z - z_0}{c} \end{cases}$$

For each equation in the above system, the parameter t must be equal, yieldinf the symmetric equation of a line:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

2.4 Finding Intersection

Suppose we have two lines with the following parametric equations:

$$\begin{cases} x_1(t_1) = a_1 t_1 + x_{10} \\ y_1(t_1) = b_1 t_1 + y_{10} \\ z_1(t_1) = c_1 t_1 + z_{10} \end{cases} \qquad \begin{cases} x_2(t_2) = a_2 t_2 + x_{20} \\ y_2(t_2) = b_2 t_2 + y_{20} \\ z_2(t_2) = c_2 t_2 + z_{20} \end{cases}$$

These two lines intersect exactly where $(x_1, y_1, z_1) = (x_2, y_2, z_2)$.

$$\begin{cases} a_1 t_1 + x_{10} = a_2 t_2 + x_{20} \\ b_1 t_1 + y_{10} = b_2 t_2 + y_{20} \\ c_1 t_1 + z_{10} = c_2 t_2 + z_{20} \end{cases}$$

We know the value of a, b, c, x_0, y_0, z_0 for both equations. There are two variables to solve: t_1 and t_2. But this is an overdetermined system, with three linear equations for two variables, so there is only a solution (and intersection) if these equations are not independent. If there is, then we can use t_1 and t_2 to find the point (or infinity points) of intersection.

3 Planes

The equation of a plane in 3D passing through (x_0, y_0, z_0) is:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

$$\begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$$

Alternative, expanding the first equation, there exists a scalar value d such that the plane is described by:

$$ax + by + cz + d = 0$$

Subtracting by x_0, y_0, z_0 shifts the plane. The plane parallel to the original plane, but which passes through the origin, is given by $(x, y, z) \cdot (a, b, c) = 0$. Which means that for all points on the plane, (a, b, c) is orthogonal to it. Hence $n = (a, b, c)$ is orthogonal to the plane.

3.1 Finding Intersection

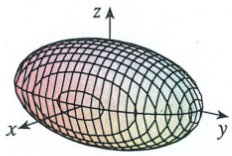
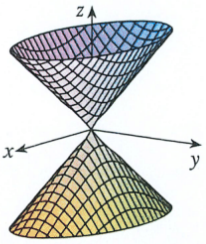
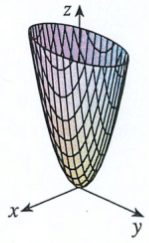
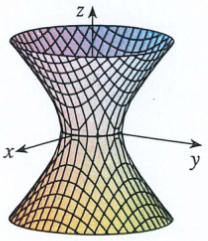
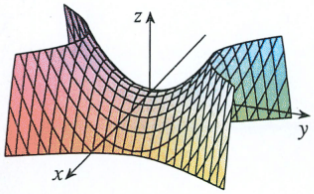
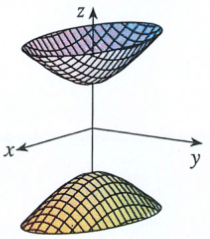
Two planes intersect if and only if the vectors normal to the plane point in different direction and are not parallel.

$$a_1x + b_1y + c_1z + d_1 = 0$$
$$a_2x + b_2y + c_2z + d_2 = 0$$

A point on the intersection of these two planes satisfy both equations. From here, set any variables (x, y, z) to a parameter t and substitute. Then use elimination to solve for the two remaining variables in terms of t (DO NOT ELIMINATE t). Now we have a parametric system describing the line of intersection.

4 Quadric Surfaces

Table 1 Graphs of Quadric Surfaces

Surface	Equation	Surface	Equation
Ellipsoid 	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ All traces are ellipses. If $a = b = c$, the ellipsoid is a sphere.	**Cone** 	$\frac{z^2}{c^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ Horizontal traces are ellipses. Vertical traces in the planes $x = k$ and $y = k$ are hyperbolas if $k \neq 0$ but are pairs of lines if $k = 0$.
Elliptic Paraboloid 	$\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$ Horizontal traces are ellipses. Vertical traces are parabolas. The variable raised to the first power indicates the axis of the paraboloid.	**Hyperboloid of One Sheet** 	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ Horizontal traces are ellipses. Vertical traces are hyperbolas. The axis of symmetry corresponds to the variable whose coefficient is negative.
Hyperbolic Paraboloid 	$\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$ Horizontal traces are hyperbolas. Vertical traces are parabolas. The case where $c < 0$ is illustrated.	**Hyperboloid of Two Sheets** 	$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ Horizontal traces in $z = k$ are ellipses if $k > c$ or $k < -c$. Vertical traces are hyperbolas. The two minus signs indicate two sheets.